Weather Prediction using Machine Learning

1. Problem Statement

The objective of this project is to build a machine learning model that can predict house prices based on various features such as area, number of bedrooms, bathrooms, parking availability, and furnishing status.

This is a regression problem, where the goal is to predict a continuous numerical output, i.e., the price of a house.

The model aims to learn the relationship between input features (independent variables) and the house price (dependent variable) so that it can accurately estimate the price of new or unseen properties. This can be useful for buyers, sellers, and real estate agents in making informed decisions.

2. Import Libraries

```
import pandas as pd
import numpy as np
```

3. Load Dataset

```
from google.colab import files
uploaded = files.upload()

df = pd.read_csv("house_price_50k.csv")
df.head()
```

Choose Files house_price_50k.csv

house_price_50k.csv(text/csv) - 4229798 bytes, last modified: 4/20/2026 - 100% done

```
Saving house_price_50k.csv to house_price_50k.csv
```

	area	bedrooms	bathrooms	floors	age	distance	garage	parking	garden	sec
0	1360	6	2	2	9	10	0	0	1	
1	4272	5	2	3	24	8	1	0	0	
2	3592	1	4	3	4	20	0	1	1	
3	966	5	2	2	6	14	1	0	0	
4	4926	4	3	1	18	9	0	0	0	

Next steps: Generate code with df New interactive sheet

4. Data Preprocessing

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = le.fit_transform(df[col])

df.head()
```

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Next steps: Generate code with df New interactive sheet

5. Create Target Variable

```
X = df.drop('price', axis=1)   # Features (independent variables)
y = df['price']               # Target (dependent variable)
```

6. Define Features & Target

```
X = df.drop('price', axis=1)   # Features
y = df['price']                # Target

y.head()
```

	price
0	5.952493e+05
1	1.571208e+06
2	1.379943e+06
3	4.367819e+05
4	1.792425e+06

dtype: float64

```
df['Price_Category'] = pd.cut(
    df['price'],
    bins=[0, 3000000, 6000000, 10000000],
    labels=['Low', 'Medium', 'High']
)

df[['price', 'Price_Category']].head()
```

	price	Price_Category
0	5.952493e+05	Low
1	1.571208e+06	Low
2	1.379943e+06	Low
3	4.367819e+05	Low
4	1.792425e+06	Low

7. Train-Test Split

```
X = df.drop('price', axis=1)
y = df['price']
```

8. Train Model

```
X = df.drop('price', axis=1)
y = df['price']
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
df = df.drop('Price_Category', axis=1)
```

```
df = pd.get_dummies(df, drop_first=True)
```

```
X = df.drop('price', axis=1)
y = df['price']
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

9. Accuracy

```
from sklearn.tree import DecisionTreeRegressor

model = DecisionTreeRegressor(random_state=42)
model.fit(X_train, y_train)
```

```
▾ DecisionTreeRegressor
DecisionTreeRegressor(random_state=42)
```

10. ML Life Cycle

The machine learning life cycle followed in this project includes:

Data Collection

Data Preprocessing

Exploratory Data Analysis

Feature Selection

Model Training

Model Testing

Model Evaluation

11. Final Conclusion

A Decision Tree model was successfully built to predict weather conditions. The model achieved a satisfactory accuracy, indicating that the dataset and selected features are suitable for prediction. The project demonstrates the complete machine learning workflow from data preprocessing to model evaluation.